

Editorial

In the oil industry, analytical chemistry is facing new challenges as the sector adapts to the energy transition and the growing emphasis on sustainable development.

Traditional fuels still demand precise and reliable analysis to optimize efficiency, reduce costs, and meet increasingly stringent environmental regulations. At the same time, emerging fields such as carbon capture and biofuel integration are driving the need for advanced research and innovation.

Within this context, foams are proving to be an effective solution in gas mobility control, enhanced oil recovery (EOR), and carbon capture, utilization, and storage (CCUS) applications.

More than ever, foam analysis has become an essential tool, offering unique insights into foam stability, texture, and lifetime under realistic conditions.

This newsletter highlights the key features of the FOAMSCAN™ High Temperature–Mid Pressure instrument and the benefits it brings to EOR and CCUS research applications.

The TECLIS Team



FOAMSCAN™
Foam analyzer & defoamer tester



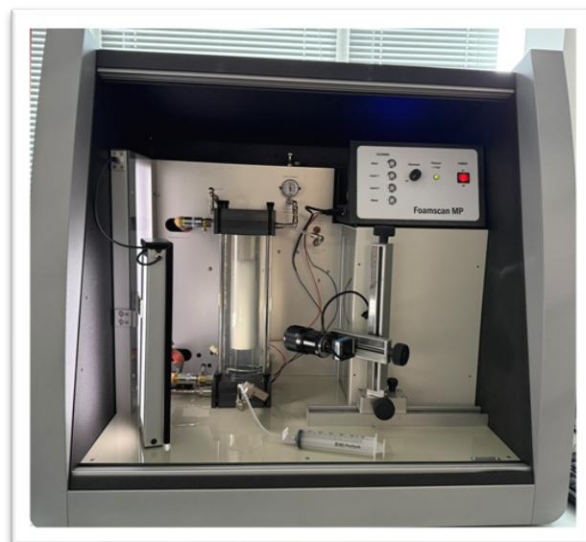
TRACKER™
Optical Drop Tensiometer &
Interfacial rheometer

FOAMSCAN™ High Temperature-Mid Pressure

Study liquid foams at temperatures up to 120 °C and pressures up to 8 bar

For many researchers, analyzing foams under pressure remains a challenge, often requiring complex and costly laboratory equipment.

The **FOAMSCAN™ HTMP** provides a practical and science-based solution, enabling foam analysis at temperatures up to 120 °C and pressures up to 8 bar (116 psi).



FOAMSCAN™ HTMP equipped with mid-pressure double-walled Cylindrical Glass Measuring tube

With a smart modular architecture, the **FOAMSCAN™ HTMP** can be configured with:

- A mid-pressure, double-walled cylindrical glass measuring tube for measurements up to 120 °C / 8 bar,
- Or a standard double-walled cylindrical glass measuring tube, equipped with electrodes and prisms, offering the full range of features and benefits of the standard **FOAMSCAN™** at Atmospheric pressure and temperature up to 90°C.

Both measuring tubes are easily exchanged thanks to quick connectors to switch from standard to mid-pressure measurements.



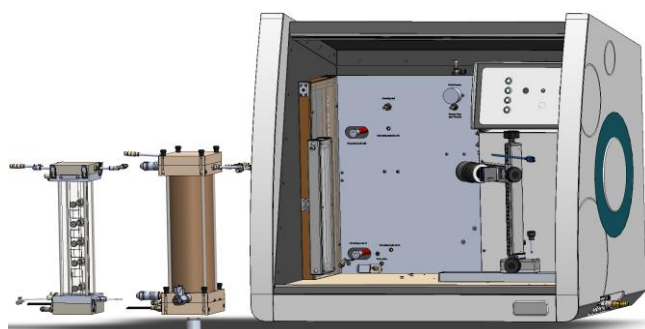
100% Software-Controlled, measurements start automatically once both pressure and temperature settings are reached.

The FOAMSCAN™ HTMP combines image analysis with conductivity measurements, delivering an exhaustive and meaningful set of results.

	Foamability	Foam wetness	Antifoam effectiveness	Foam Decay	Drainage	Ostwald ripening	Coalescence
Foam volume (mL)	•		•	•			
Liquid Volume (mL)	•	•		•	•		
Liquid Fraction (%)		•		•	•		
Volume of gas injected	•						
Foam capacity	•						
Foam expansion	•						
Foam density	•	•		•	•		
Bikerman Index (sec.)	•						
Foam volume half-life time (sec)			•	•			
Liquid stability half-life time (sec)	Mid-Pressure Tube						
Foam local liquid fraction (%)		•		•	•		
Bubbles Count (number)				•		•	•
Bubbles size (mm)						•	
Polydispersity Index						•	•
Liquid fraction (%) image analysis	Standard Tube						

The automatic cleaning system works for both standard and mid-pressure measuring double-walled glass tubes, eliminating the need to dismantle the setup for manual cleaning.

In addition, the FOAMSCAN™ HTMP can generate and analyze foams created by depressurization and be adapted to a version fully compatible with HCl solutions, expanding its range of applications.



FOAMSCAN™ HTMP equipped with standard and mid-pressure measuring double-walled glass tubes

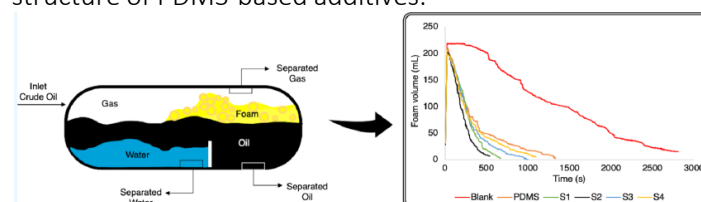
Applications

The following two scientific papers show how the FOAMSCAN™ HTMP delivers reliable and meaningful data whether at mid pressure and temperature or in standard lab settings, supporting research in enhanced oil recovery (EOR) and carbon capture, utilization, and storage (CCUS).

Physio-Chemical Analysis of Amide and Amine Poly(dimethylsiloxane) - Modified Defoamer for Efficient Oil–Water Separation

Almila Hassan et al., ACS Omega 2021

The study investigates poly(dimethylsiloxane) (PDMS)-modified defoamers, functionalized with amide and amine groups, to enhance oil–water separation. The objective was to improve demulsification and defoaming efficiency in petroleum processing by tailoring the molecular structure of PDMS-based additives.



The FOAMSCAN™ HTMP played a pivotal role in this research by enabling:

- to quantify foam generation and decay under controlled pressure and temperature conditions, replicating real oilfield environments.
- precise monitoring of foam volume, bubble size, and half-life at pressures relevant to petroleum systems.
- dynamic measurements of foam collapse with high sensitivity.

Results showed that amide-modified PDMS collapsed foams more rapidly under pressure, confirming its superior efficiency in realistic operating conditions.

The FOAMSCAN™ HTMP validated that the designed defoamers not only work in ambient lab tests but also retain effectiveness under reservoir-like conditions; a crucial step for field applicability.

<https://doi.org/10.1021/acsomega.1c00350>



Synergy of Fly Ash and Surfactant on Stabilizing CO₂/N₂ Foam for CCUS in Energy Applications

Jabir Dubaish Raib et al., Energies (2025)

This research addresses the challenge of stabilizing foams for enhanced oil recovery (EOR) and carbon capture, utilization, and storage (CCUS). Pure CO₂ foams are typically unstable due to high gas solubility and weak interfacial interactions, while N₂ foams are more stable but less effective for CCUS. To overcome these limitations, the study explored blended CO₂/N₂ foams, mimicking flue gas, stabilized with surfactants (SDS, SDBS, CTAB) in combination with nano fly ash and clay particles.

The findings not only improve the efficiency of EOR processes but also highlight a cost-effective CCUS strategy, while valorizing fly ash waste as a sustainable additive.

<https://doi.org/10.3390/en18154181>



Coming Events

Meet us at TAD 2025

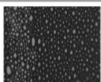
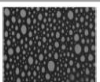
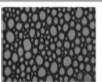
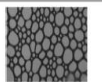
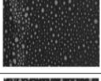
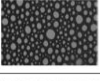
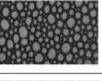
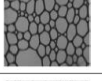
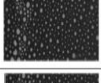
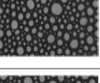
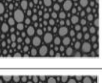
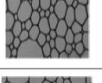
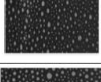
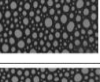
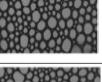
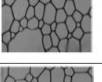
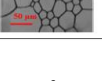
Transition Analytics for Decarbonization

The first edition of the TAD Conference will take place from October 1–3, 2025, at the CNRS headquarters in Paris.

The TECLIS Team will present results on the *Influence of CO₂ Content on the Formation, Stability, and Structural Evolution of Liquid Foams*.



<https://tad-conference.sciencesconf.org/>

Time (min)	0	5	10	20
100% N ₂				
75% N ₂ + 25% CO ₂				
50% N ₂ + 50% CO ₂				
25% N ₂ + 75% CO ₂				
100% CO ₂				

Analysis of Foam Size and Structure

Table 3. Visual micrographs showing foam structure evolution of 0.8 wt % SDS with different gas ratios at different times after 20 min

Experiments were performed at ambient temperature and pressure. Although the high-pressure capabilities of the equipment were not employed, the FOAMSCAN™ HTMP, equipped with the standard double-walled cylindrical glass measuring tube, transformed the study from simple stability tests into a detailed structural analysis by enabling:

- Real-time monitoring of foam decay using image analysis and conductivity measurements.
- Statistical analysis of bubble size and distribution over time, capturing coalescence and coarsening events.
- Comparison of foam stability across different CO₂/N₂ ratios and surfactant concentrations, with high accuracy.
- Correlating bubble size with foam half-life and stability.

